

eGov Agent | Agentic orchestration for eGovPH

eGovPH already provides the digital front door and shared government rails. eGov Agent adds a consent-aware execution layer that plans, coordinates, verifies, and documents multi-agency services for the citizen.

1 PROTOTYPE REALITY

STATUS	HONEST BOUNDARY
Implemented UX	Chat, action trace, uploads, vault/memory interfaces, file stamps, printable results
Simulated	National ID/eVerify, agency records, appointments, eGovPay/eReceipt, eReport dispatch
Not yet connected	Production backend, database, workflow queue, agency credentials, real payment callbacks

The prototype proves a cohesive citizen experience; production credibility depends on explicit adapters, authoritative read-back, privacy engineering, and agency ownership of legal outcomes.

2 PRODUCTION ARCHITECTURE

Channels → API gateway → Agent orchestrator → Policy + consent → Versioned adapters → eVerify / eGovDX / eGovPay / agency-LGU systems

- Durable workflow engine and event bus isolate retries, timeouts, compensation, and human handoff from the chat session.
- Trusted services provide minimal memory, encrypted vault storage, append-only audit, observability, and revocable consent receipts.
- Each agency adapter is versioned, least-privilege, idempotent, contract-tested, rate-limited, and owned jointly with the system-of-record team.

3 TRANSACTION INVARIANTS

- Authenticate with government-approved identity; collect only the attributes required for the transaction.
- Translate intent into a bounded plan; show agencies, data, fees, and consequences before consent.
- Execute server-side with idempotency keys, bounded retries, and compensating actions.
- Read back from the authoritative system. A successful UI or model response is never proof of completion.
- Issue an agency receipt/reference; reconcile every side effect and preserve an auditable decision trail.

CORE RULE The model may propose and explain. Policy gates authorize. Government systems execute and attest.

4 MEMORY, VAULT & PRIVACY

- Use relational storage for sourced facts, a small graph for durable relationships, and vector retrieval only for user-approved unstructured context; do not make a knowledge graph the primary truth store.
- Every remembered fact carries provenance, purpose, confidence, retention, and an edit/delete path. Sensitive documents remain encrypted objects, not prompt history.
- Complete a Privacy Impact Assessment under the Data Privacy Act and NPC Circular 2023-06 before production personal-data processing.

5 eGOVPAY → VERIFIED eRECEIPT

- Agency service creates the order server-side and locks merchant, fee, currency, citizen reference, and expiry.
- Citizen is redirected to hosted eGovPay; the agent never stores card or wallet credentials.
- Backend validates signed callback or queries status server-to-server; reject mismatched amount/order/merchant and replayed events.
- Agency ledger posts exactly once using an idempotency key, then the system-of-record is read back.
- Only the agency/eReceipt service issues the authoritative receipt ID; the agent renders and stores the verified proof.

6 eREPORT WITH HUMAN ACCOUNTABILITY

- AI structures citizen narrative, geolocation, media, urgency indicators, and routing suggestions; the citizen confirms before submission.
- Dispatch fan-out is idempotent. Status comes only from agency acknowledgements, with responder ownership and escalation visible to the citizen.
- Safety-critical classifications, suspected abuse, and uncertain jurisdiction route to trained personnel; no model may claim dispatch or resolution on its own.

7 SECURITY & GOVERNANCE BASELINE

- OIDC/OAuth 2.1, ABAC, short-lived tokens, mTLS for service links, managed KMS/HSM, field/object encryption, secret rotation, malware scanning, and strict upload limits.
- Prompt-injection defenses treat documents and external text as untrusted data; tools require schemas, allowlists, policy checks, transaction scopes, and human approval for high-impact actions.
- Append-only audit records who/what acted, consent basis, adapter/version, hashes, result, and correlation ID—without duplicating unnecessary personal data.

8 PILOT, COST & SUCCESS GATES

WINDOW	DELIVERABLE
0-6 weeks	Governance, PIA, service blueprints, adapter contracts
6-12 weeks	Read-only discovery, identity, consent, sandbox integrations
12-24 weeks	Transactional pilot: payment, eReceipt, reporting, human escalation
6-12 months	2 national agencies + 1 LGU; hardening and scale-out

Illustrative 9-12 month pilot: PHP 18.0M-33.5M for up to 50,000 monthly active users. Validate through agency scoping and government procurement; this is not a vendor quotation.

- ≥95% correct service routing; zero duplicate irreversible effects in acceptance tests; ≥99.95% payment reconciliation.
- Zero unresolved critical security/privacy findings at go-live; WCAG 2.2 AA; full audit reconstruction for sampled transactions.

OFFICIAL BASIS

eGovPH Core (core.e.gov.ph) · eGovPay (egovpay.gov.ph) · PSA National ID eVerify (psa.gov.ph; everify.gov.ph) · Data Privacy Act and NPC Circular 2023-06 (privacy.gov.ph) · RA 11032 (lawphil.net)

RECOMMENDATION Approve a tightly scoped, sandbox-first pilot. Keep agencies authoritative; make consent, verification, reconciliation, and human escalation non-negotiable platform capabilities.